INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location	ND facility ND_way_1422191477 -- station KABR, America/Chicago
Committed pair	OSM 1422191475 -> 1422191477 OSM way 1422191475 / OSM way 1422191477
Facade gap	102.5 m between the two halls
Weather record	43,427 real hourly records from KABR
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.2421 C at 60 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2022-10-19 (crossing)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	1 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 6 of 24 hours with 1 mode change. The reactive incumbent took 17 hours -- 11 more -- but declared 2 unsafe hours against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 16 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	6 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	17 of 24 hours free cooling, 2 mode change(s), 2 unsafe hour(s) the incumbent broke its own switch budget 1 time(s) to stay safe 16 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	-5.723	18.0	-6.110	1.242
01	mechanical	yes	switch budget	-6.044	18.0	-6.720	1.085
02	mechanical	yes	switch budget	-6.273	18.0	-7.779	1.261
03	mechanical	yes	switch budget	-6.602	18.0	-8.280	1.173
04	mechanical	yes	switch budget	-6.232	18.0	-7.220	1.175
05	mechanical	yes	switch budget	-6.512	18.0	-6.720	1.132
06	mechanical	yes	switch budget	-6.677	18.0	-7.094	1.199
07	mechanical	yes	switch budget	-5.592	18.0	-7.654	1.504
08	mechanical	yes	switch budget	-2.425	18.0	-5.000	2.239
09	mechanical	yes	switch budget	0.617	18.0	-0.610	2.477

10	mechanical	yes	switch budget	2.257	18.0	2.220	2.542
11	mechanical	yes	switch budget	5.866	18.0	7.220	2.300
12	mechanical	yes	switch budget	8.917	18.0	10.560	1.832
13	mechanical	yes	switch budget	11.737	18.0	14.440	1.700
14	mechanical	yes	switch budget	15.075	18.0	17.780	1.355
15	mechanical	yes	switch budget	17.038	18.0	19.440	1.322
16	mechanical	no	dry-bulb	18.686	18.0	20.560	1.090
17	mechanical	no	dry-bulb	18.931	18.0	18.890	1.145
18	FREE	yes	-	17.010	18.0	14.440	1.246
19	FREE	yes	-	14.519	18.0	8.890	1.480
20	FREE	yes	-	15.287	18.0	11.670	1.855
21	FREE	yes	-	12.598	18.0	10.560	2.081
22	FREE	yes	-	9.528	18.0	10.000	1.731
23	FREE	yes	-	10.636	18.0	8.890	1.562

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -5.723 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -6.044 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -6.273 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -6.602 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -6.232 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -6.512 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -6.677 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -5.592 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -2.425 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.617 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.257 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.866 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.917 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

13:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.737 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

14:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.075 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

15:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.038 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.686 C against a 18.0 C limit, so it fails by 0.686 C. It would take a limit 0.686 C higher, or a bound 0.686 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.931 C against a 18.0 C limit, so it fails by 0.931 C. It would take a limit 0.931 C higher, or a bound 0.931 C tighter, to change this hour.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.010 C, which is 0.990 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.246 C, and the margin is measured, not chosen: 1.176 C of group-conditional forecast error for this hour of day, plus 0.0700 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.519 C, which is 3.481 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.480 C, and the margin is measured, not chosen: 1.411 C of group-conditional forecast error for this hour of day, plus 0.0687 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.287 C, which is 2.713 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.855 C, and the margin is measured, not chosen: 1.749 C of group-conditional forecast error for this hour of day, plus 0.1069 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.598 C, which is 5.402 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.081 C, and the margin is measured, not chosen: 1.983 C of group-conditional forecast error for this hour of day, plus 0.0974 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.528 C, which is 8.472 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.731 C, and the margin is measured, not chosen: 1.648 C of group-conditional forecast error for this hour of day, plus 0.0829 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.636 C, which is 7.364 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.563 C, and the margin is measured, not chosen: 1.481 C of group-conditional forecast error for this hour of day, plus 0.0812 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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